A CASE-STUDY BASED REPORT

ON

Loan Approval Prediction Using Ensemble Learning: A Case Study of Random Forest, XGBoost and LightGBM with SHAP Explainability

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MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Abstract

Granting a loan is one of the most consequential decisions a lender makes: approving a borrower who later defaults causes direct financial loss, while rejecting a creditworthy applicant means lost business and reduced financial inclusion. This case study presents an automated credit-assessment framework that predicts loan-approval decisions from applicant financial data using an ensemble of three tree-based learners: Random Forest, XGBoost and LightGBM.

Using a publicly available loan-application dataset of 4,269 records and 12 columns (11 applicant and loan attributes plus the decision), the data was cleaned (no missing values and no duplicates were found), explored visually and statistically, and enriched with three domain-engineered financial ratios: a loan-to-income ratio (the debt-to-income measure), a loan-to-asset ratio (the credit-utilisation measure) and a payment-to-income ratio. The moderate class imbalance (62.2% approved, 37.8% rejected) was handled with class weights, and the data was split 80/20 using stratified sampling. The three base estimators were trained individually and then combined by soft voting, which averages their predicted probabilities.

Because wrongly approving a risky applicant is far costlier than wrongly rejecting a good one, the models were evaluated with ROC-AUC and the F2-score (which weights recall more heavily than precision), in addition to accuracy, precision and recall, with the rejected (high-risk) class treated as the positive class. On the held-out test set of 854 applications the soft-voting ensemble achieved an accuracy of 100%, an F2-score of 1.000 and a ROC-AUC of 1.000, matching Random Forest and slightly improving on XGBoost and LightGBM, which each misclassified one application. SHAP analysis showed that the CIBIL credit score is by far the most influential feature and explained individual decisions at the applicant level. The near-perfect scores indicate that this dataset is almost fully separable by credit score, so they should not be expected to carry over to real lending data. The study is intended for learning purposes and is not a substitute for a lender's credit-risk process.

Keywords: loan approval prediction, ensemble learning, Random Forest, XGBoost, LightGBM, soft voting, class imbalance, F2-score, SHAP, credit risk.

Introduction

Background

Banks and financial institutions receive a large number of loan applications every day. Each application must be assessed for the applicant's ability and willingness to repay, which is traditionally judged by combining indicators such as income, employment type, credit score, assets, loan amount and loan term. No single indicator is conclusive, and the relationships between them are often non-linear: for example, a high income may still be risky when it is accompanied by a very large loan or a poor credit history.

Machine learning (ML) offers a data-driven way of learning such combined patterns from historical applications whose outcomes are already known. In supervised classification, an algorithm is trained on past applications labelled as approved or rejected and then predicts the decision for new, unseen applicants. Ensemble learning goes one step further by combining several different models so that the strengths of one compensate for the weaknesses of another, which usually gives more accurate and more stable predictions than any single model.

Motivation for the Project

The motivation for this project comes from four observations:

- Manual credit assessment is slow, inconsistent between officers and difficult to scale, whereas an automated model gives fast and uniform decisions.
- Defaults are costly. A model that is tuned to catch risky applications directly protects the lender, so the evaluation must reflect this asymmetric cost instead of relying on accuracy alone.
- Lending data is imbalanced and contains complex financial patterns, which makes it a good test-bed for tree-based ensemble methods and for imbalance-handling techniques.
- Financial regulation requires lenders to explain their decisions to applicants, so the model must be interpretable at the level of the individual applicant and not only accurate.

Overview of the Work Carried Out

The work was carried out in Python with pandas, NumPy, Matplotlib, Seaborn, scikit-learn, XGBoost, LightGBM and SHAP. The main stages were:

- **Data cleaning:** checking the shape, data types, missing values, duplicates and outliers, and removing stray spaces from the column names and text values.
- **Exploratory data analysis:** summary statistics, class balance, distributions, relationships between key features and the loan decision, and a correlation heatmap.
- **Feature engineering:** creation of the loan-to-income ratio, loan-to-asset ratio and payment-to-income ratio as domain-driven financial metrics.
- **Pre-processing:** encoding of the two categorical attributes, a stratified 80/20 train-test split and class-weight handling of the class imbalance.

- **Modelling:** training Random Forest, XGBoost and LightGBM classifiers and combining them into a soft-voting ensemble.
- **Evaluation:** accuracy, precision, recall, F2-score and ROC-AUC, together with confusion matrices, ROC curves and 5-fold cross-validation on the training data.
- **Interpretability:** SHAP analysis to show which features drive the decision globally and for an individual applicant.

Scope and Responsible-Use Statement

The scope of this study is limited to binary classification (loan approved or rejected) on a single dataset using three tree-based algorithms and one ensembling strategy, soft voting. It does not include deep learning, extensive hyper-parameter optimisation, deployment in a live lending system, external validation or any assessment by a regulator.

Responsible use: the models in this report are built for academic learning only. They must not be used to make real lending decisions. Historical lending data can contain social biases, and a model trained on it may reproduce them, so any real credit decision must remain under the supervision of qualified credit professionals and applicable law. The dataset used contains no personally identifiable information.

Problem Statement

Case-Study Scenario

Consider a retail bank that processes thousands of loan applications every month. For each applicant the bank records the number of dependents, education level, whether the applicant is self-employed, annual income, the requested loan amount and loan term, the applicant's CIBIL credit score and the value of the applicant's residential, commercial, luxury and bank assets. Loan officers have limited time, and wrongly approving an applicant who later defaults is far more expensive than asking a good applicant for additional documents. The bank would therefore like an automated tool that predicts whether an application should be approved, flags high-risk applications for closer review, and gives a clear reason for each decision.

Central Case-Study Question

Can an ensemble of Random Forest, XGBoost and LightGBM, trained on applicant financial data enriched with domain-engineered ratios, predict loan-approval decisions more reliably than any of the individual models, while remaining explainable for every applicant?

Formal Problem Definition

Let $X = (x_1, x_2, \ldots, x_{14})$ denote the vector of attributes of a loan application, including the engineered loan-to-income, loan-to-asset and payment-to-income ratios, and let $y \in \{0, 1\}$ denote the decision, where $y = 1$ means the application is rejected (high risk) and $y = 0$ means it is approved. Given a training set of labelled applications $\{(X_i, y_i)\}$, the task is to learn base functions f_RF(X), f_XGB(X) and f_LGBM(X) that each output the probability that the application is high risk, and to combine them by soft voting into an ensemble probability $P(y = 1 \mid X) = (f_RF + f_XGB + f_LGBM) / 3$. The final prediction $\hat{y}$ is obtained by applying a decision threshold of 0.5 to this probability, and the quality of the prediction is measured on a held-out test set using accuracy, precision, recall, F2-score and ROC-AUC.

Objectives that Guide the Study

- To clean and validate the loan dataset and make it suitable for modelling.
- To explore the data and identify which applicant attributes are most associated with the loan decision.
- To engineer domain-driven financial metrics (loan-to-income, loan-to-asset and payment-to-income ratios) that capture an applicant's repayment burden.
- To address class imbalance so that the models are not biased towards the majority class.
- To build Random Forest, XGBoost and LightGBM classifiers and combine them with a soft-voting ensemble.
- To evaluate the models with ROC-AUC and the F2-score, which prioritise the reduction of costly default risk, alongside the standard metrics.
- To use SHAP to provide applicant-level explanations that support regulatory transparency.

Assumptions and Limitations

- The dataset is assumed to be a representative and correctly labelled sample of past loan decisions; the Approved/Rejected label is taken as the ground truth and the Rejected class is treated as the positive (high-risk) class.
- The dataset is small (4,269 records), so the test set contains 854 applications; one application changes the accuracy by about 0.12 percentage points.
- A single train-test split (random_state = 42) was used for the final comparison, supported by 5-fold stratified cross-validation on the training data.
- The dataset has no columns for existing debt or credit limits, so the debt-to-income and credit-utilisation measures were approximated by the loan-to-income and loan-to-asset ratios computed from the available columns.
- Some residential asset values are slightly negative (28 records, minimum −100,000); they were retained as they are.
- Historical decisions may contain human or social bias, which a model trained on them can learn.
- Models were trained with fixed, standard hyper-parameters and were not tuned, and no external validation dataset was used.

Dataset and Exploratory Data Analysis

About the Dataset

The study uses a loan-application dataset stored as loan_approval_dataset.csv, a publicly available dataset widely used for ML education. It contains 4,269 loan applications and 13 columns: an identifier (loan_id), 11 applicant and loan attributes, and one target column (loan_status) recording whether the loan was approved or rejected. The loan_id column carries no predictive information and was dropped, leaving 12 columns. No missing values and no duplicate records were present. Table 1 describes each attribute.

Table 1: Attributes of the loan dataset

Attribute	Description	Type
no_of_dependents	Number of dependents of the applicant (0–5)	Numeric
education	Education level (Graduate / Not Graduate)	Categorical
self_employed	Whether the applicant is self-employed (Yes / No)	Binary
income_annum	Annual income of the applicant	Numeric
loan_amount	Loan amount requested	Numeric
loan_term	Loan term in years (2–20)	Numeric
cibil_score	CIBIL credit score of the applicant (300–900)	Numeric
residential_assets_value	Value of residential assets	Numeric
commercial_assets_value	Value of commercial assets	Numeric
luxury_assets_value	Value of luxury assets	Numeric
bank_asset_value	Value of bank assets	Numeric
loan_to_income (engineered)	Loan amount ÷ annual income (debt-to-income measure)	Numeric
loan_to_assets (engineered)	Loan amount ÷ total assets (credit-utilisation measure)	Numeric
payment_to_income (engineered)	Monthly instalment (loan ÷ term in months) ÷ monthly income	Numeric
loan_status (target)	Decision: Rejected = 1 (high risk), Approved = 0	Target (binary)

The target is moderately imbalanced: 2,656 applications (62.2%) were approved and 1,613 applications (37.8%) were rejected, a ratio of about 1.65 to 1. Because of this imbalance, accuracy alone is not a reliable metric, so ROC-AUC and the F2-score are used as the main measures of performance, and the imbalance is handled with class weights instead of resampling.

Purpose of EDA

Exploratory Data Analysis (EDA) was performed before modelling to understand the structure and quality of the data, to detect anomalies, outliers and class imbalance, to see how individual

attributes relate to the loan decision, and to detect strongly correlated features. These insights guide the choice of pre-processing steps and help interpret the behaviour of the models afterwards.

Preparation Steps Actually Performed

- **Loading:** the data was loaded with pandas from loan_approval_dataset.csv and the first rows were inspected; the shape was (4269, 13). The column names and text values contained leading spaces, which were stripped.
- **Missing values:** isnull().sum() showed zero missing values in every column.
- **Duplicates:** no duplicate rows were found once loan_id was excluded.
- **Outliers and consistency:** the IQR rule found no outliers in income, loan amount, CIBIL score or luxury assets; 28 residential-asset values were slightly negative (minimum −100,000) and were retained.
- **Feature engineering:** loan_to_income = loan_amount ÷ income_annum; loan_to_assets = loan_amount ÷ (sum of the four asset values); payment_to_income = loan_amount ÷ (loan_term × income_annum), which equals the monthly instalment divided by the monthly income.
- **Encoding:** education was encoded as 1 for Graduate and 0 for Not Graduate, self_employed as 1 for Yes and 0 for No, and the target as 1 for Rejected and 0 for Approved.
- **Feature/target separation:** X contains the 14 input features (11 original and 3 engineered) and y the target column.
- **Train-test split:** 80% training (3,415 records) and 20% testing (854 records) using train_test_split with random_state = 42 and stratify = y, so that both sets keep the same class proportions (37.8% rejected).
- **Class imbalance:** class weights were used instead of resampling: class_weight = 'balanced' for Random Forest and LightGBM, and scale_pos_weight = 1.647 (approved ÷ rejected in the training set) for XGBoost.
- **Feature scaling:** not required for the tree-based models used, because decision trees are insensitive to the scale of the features.

EDA Exploration - Implementation

Descriptive statistics were first obtained with df.describe(). The average applicant has an annual income of about 5.06 million (range 0.2–9.9 million), requests a loan of about 15.1 million (range 0.3–39.5 million) for an average term of 10.9 years (range 2–20), and has an average CIBIL score of about 600 (range 300–900). Applicants have 2.5 dependents on average, 50.2% are graduates and 50.4% are self-employed.

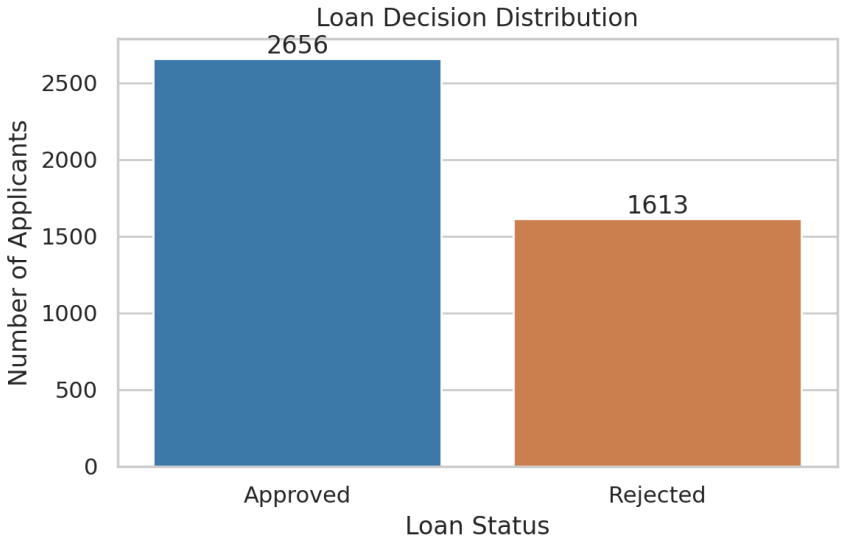


Figure 1: Distribution of the target (loan decision) classes

Figure 1 confirms the moderate imbalance: roughly 6 in 10 applications are approved.

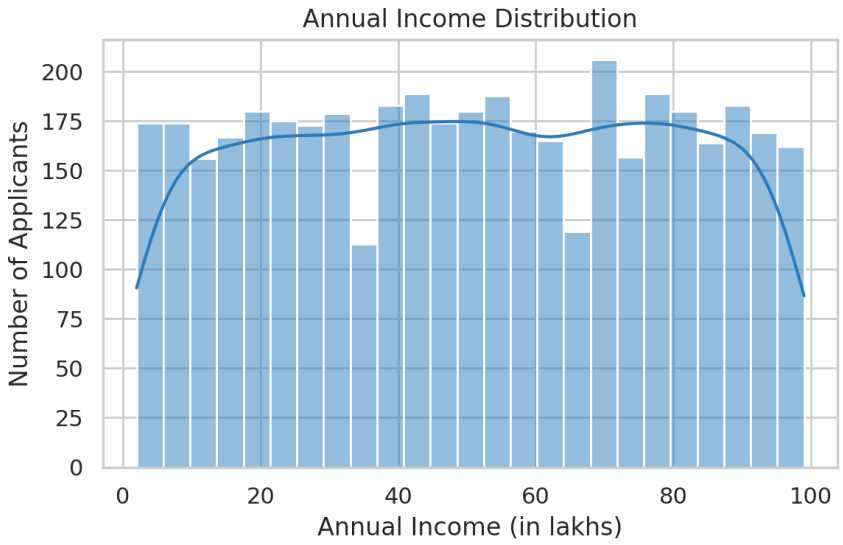


Figure 2: Annual income distribution of the applicants

The income histogram is almost flat, with applicants spread evenly between about 0.2 and 9.9 million. There is no long tail and no outlier, which is typical of a synthetic or uniformly sampled dataset.

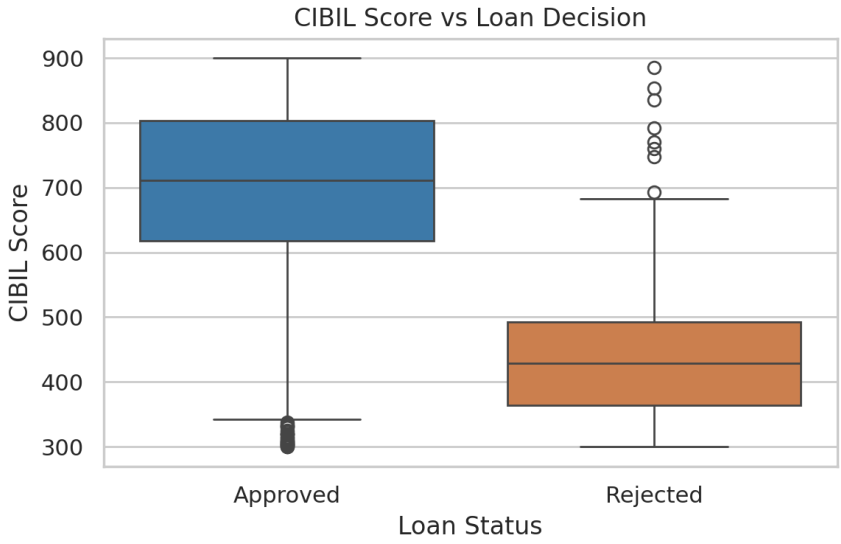


Figure 3: CIBIL score versus loan decision

Approved applicants have a much higher CIBIL score (average about 703) than rejected applicants (average about 429). The two boxes barely overlap, which is consistent with the strong negative correlation (−0.77) between CIBIL score and the rejection target. The rejection rate is 89.3% for scores up to 500, 46.9% for scores of 501–600, and only about 0.5% for scores above 600.

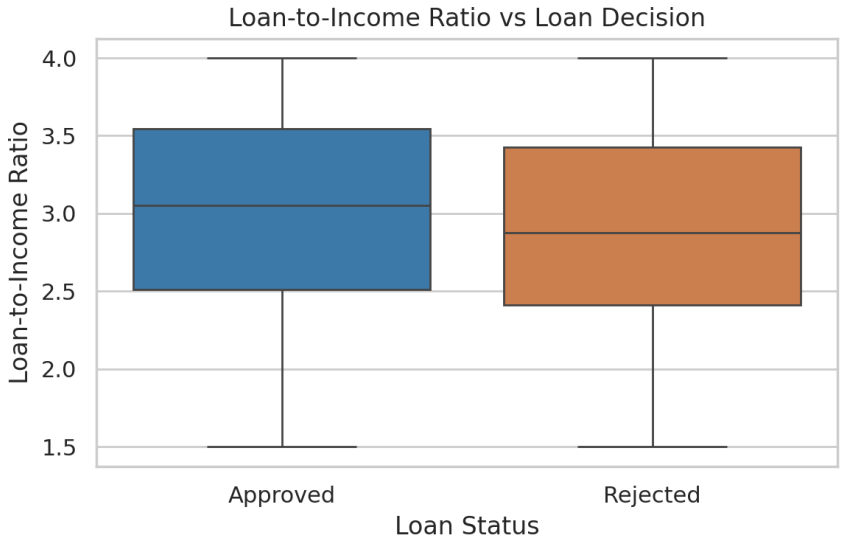


Figure 4: Loan-to-income ratio versus loan decision

The engineered loan-to-income ratio ranges from 1.5 to 4.0 and is almost identical for approved and rejected applications (averages 3.03 and 2.92). On its own it is therefore a weak predictor (correlation −0.09), and its value comes from interactions with other features such as the loan term and the CIBIL score.

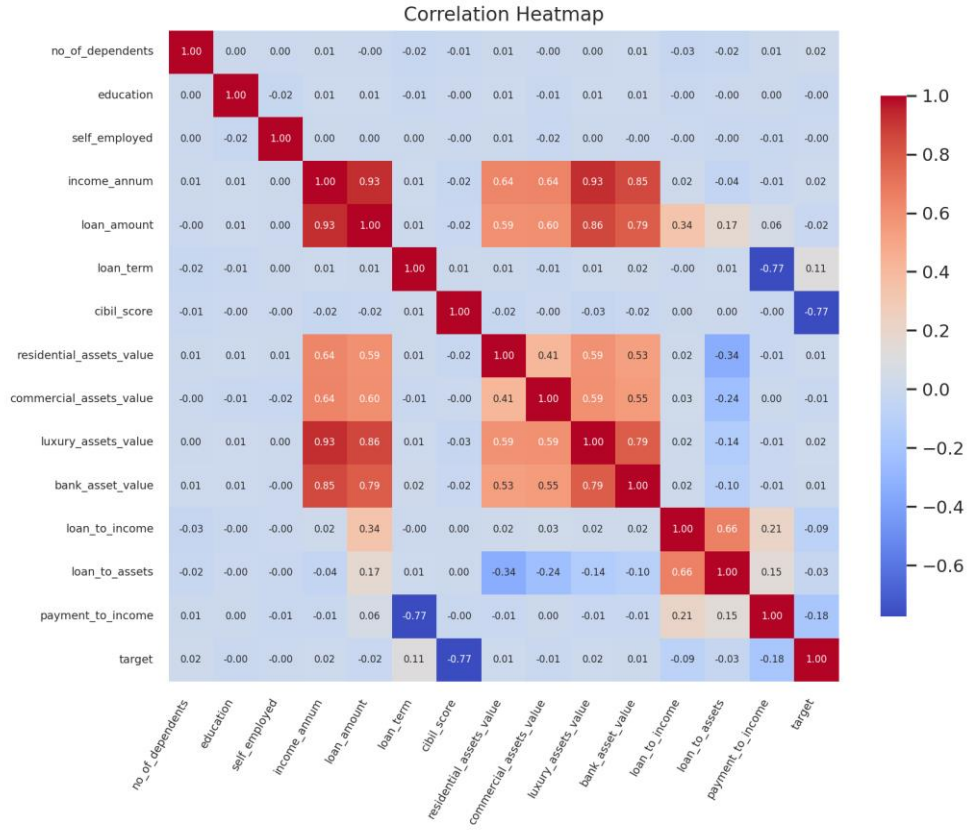


Figure 5: Correlation heatmap of all attributes

Summary of EDA Findings

- The data is clean: no missing values and no duplicates, and the target is moderately imbalanced (62.2% approved vs 37.8% rejected).
- CIBIL score is by far the strongest predictor of the decision (correlation −0.77 with rejection). A simple rule of rejecting every score below 550 would already classify about 95% of all applications correctly.
- The other correlations with the target are weak: payment_to_income (−0.18), loan_term (+0.11) and loan_to_income (−0.09); loan amount, income, assets, education, self-employment and dependents are all close to zero (below 0.02 in absolute value).
- Education and self-employment do not influence the decision: the rejection rate is 38.0% for non-graduates and 37.5% for graduates, and 37.8% for both self-employed and salaried applicants.
- There is strong correlation between the financial attributes: income and loan amount (0.93), income and luxury assets (0.93), loan amount and luxury assets (0.86), and between payment_to_income and loan_term (−0.77). This does not harm tree-based models, which handle correlated features well.
- The attributes are on very different scales (loan amounts of millions against ratios below 4), but no scaling is needed for tree-based models.

Algorithm Selection and Implementation

The problem is a binary classification task on tabular financial data with non-linear interactions between attributes. Three tree-based algorithms were selected because they are well suited to such data, need no feature scaling and each learns in a different way, which makes their combination more informative:

- **Random Forest** builds many decision trees on random subsets of the data and features and averages their votes (bagging). It is robust to noise and outliers, resists over-fitting and gives a strong, stable baseline.
- **XGBoost** builds trees one after another, where each new tree corrects the errors of the previous ones (gradient boosting) with regularisation. It captures complex non-linear patterns and has built-in support for imbalanced classes through scale_pos_weight.
- **LightGBM** is a fast gradient-boosting framework that grows trees leaf-wise and uses histogram-based splitting. It trains quickly and handles imbalanced data efficiently.
- **Soft-voting ensemble** averages the predicted probabilities of the three models and applies a decision threshold of 0.5 to the average. Compared with hard (majority) voting, soft voting keeps the confidence of each model, so a very confident model can outweigh two uncertain ones.

Table 2: Model configuration used in the implementation

Model	Library class	Settings used
Random Forest	RandomForestClassifier	n_estimators = 300, min_samples_leaf = 2, class_weight = 'balanced', random_state = 42
XGBoost	XGBClassifier	n_estimators = 300, learning_rate = 0.05, max_depth = 5, subsample = 0.8, colsample_bytree = 0.8, scale_pos_weight = 1.647
LightGBM	LGBMClassifier	n_estimators = 300, learning_rate = 0.05, num_leaves = 31, class_weight = 'balanced'
Soft-voting ensemble	VotingClassifier	voting = 'soft', equal weights, estimators = Random Forest, XGBoost, LightGBM

Implementation. All three base models were trained on the training set (3,415 records) and the soft-voting ensemble was built from them. Predictions were made on the 854 test records (323 rejected and 531 approved). Performance was measured with accuracy, precision, recall, F2-score and ROC-AUC, where the positive class (1) is Rejected, i.e. a high-risk application. The F2-score is defined as $F2 = 5 \times (\text{Precision} \times \text{Recall}) \div (4 \times \text{Precision} + \text{Recall})$, so recall is weighted four times as much as precision, which matches the lender's priority of not letting risky applications through.

Table 3: Performance of the models on the test set (854 applications)

Model	Accuracy	Precision	Recall	F2-Score	ROC-AUC
Random Forest	1.000	1.000	1.000	1.000	1.000
XGBoost	0.999	0.997	1.000	0.999	1.000
LightGBM	0.999	1.000	0.997	0.998	1.000
Soft-Voting Ensemble	1.000	1.000	1.000	1.000	1.000

The confusion matrices (Figure 6) show how many approved and rejected applications each model classified correctly on the test set.

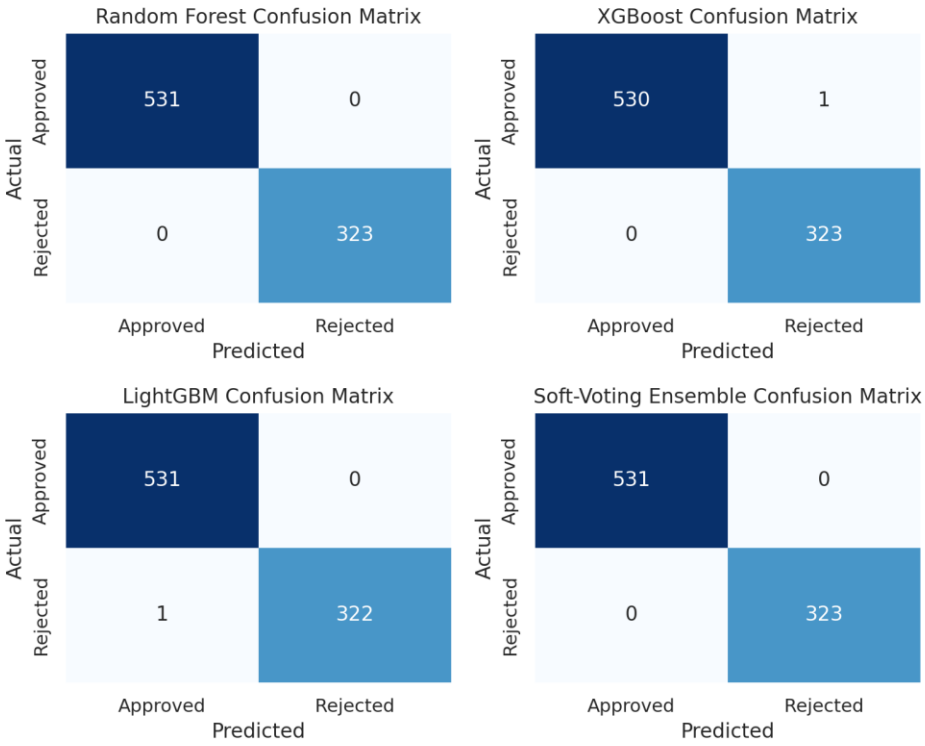


Figure 6: Confusion matrices of Random Forest, XGBoost, LightGBM and the soft-voting ensemble

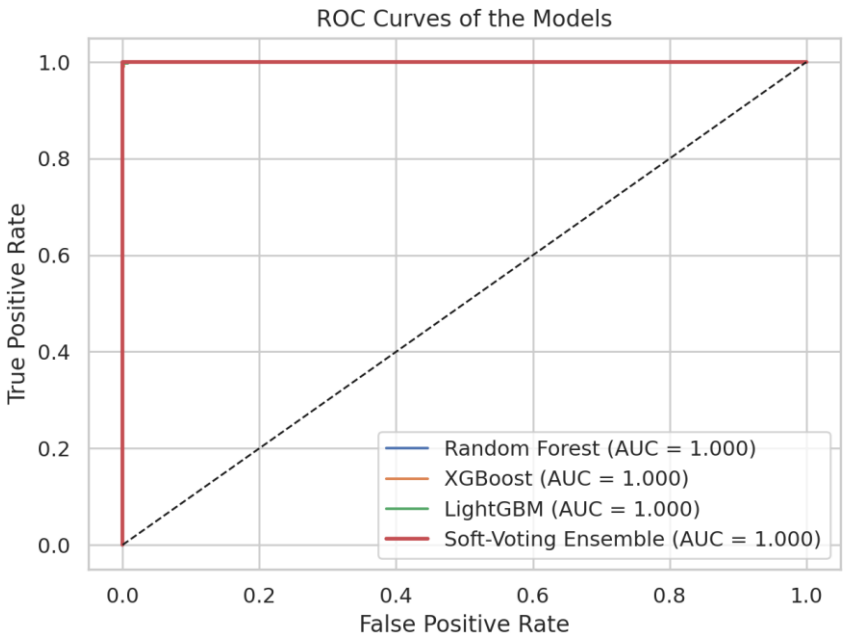


Figure 7: ROC curves of the three base models and the ensemble

Table 4: 5-fold stratified cross-validation on the training set (mean ± standard deviation)

Model	ROC-AUC	F2-Score
Random Forest	1.000 ± 0.000	0.995 ± 0.004
XGBoost	1.000 ± 0.000	0.997 ± 0.003
LightGBM	1.000 ± 0.000	0.999 ± 0.002
Soft-Voting Ensemble	1.000 ± 0.000	0.999 ± 0.002

Discussion of results. The soft-voting ensemble and Random Forest classified all 854 test applications correctly (accuracy, precision, recall, F2-score and ROC-AUC all equal to 1.000). XGBoost wrongly rejected one good applicant (precision 0.997), and LightGBM wrongly approved one risky applicant (recall 0.997), so the ensemble removes the single mistake made by each of these two models, but it only ties with Random Forest and does not beat it. The cross-validation results in Table 4 agree with the test results: ROC-AUC is 1.000 in every fold, and the cross-validated F2-score of the ensemble (0.999) is on par with LightGBM (0.999) and higher than Random Forest (0.995) and XGBoost (0.997). Because wrongly approving a risky applicant is the costly error, recall of the rejected class was given more weight than accuracy; all four models reach a recall of at least 0.997.

These near-perfect scores should be interpreted with care. The EDA showed that the CIBIL score alone separates the classes almost completely, and the differences between the models amount to one application out of 854. The dataset therefore appears to be rule-driven or synthetic, and real lending data, where the decision depends on noisier and more overlapping factors, would give lower and more varied scores. The result demonstrates the workflow and the benefit of ensembling for stability, not a ready-to-use credit model.

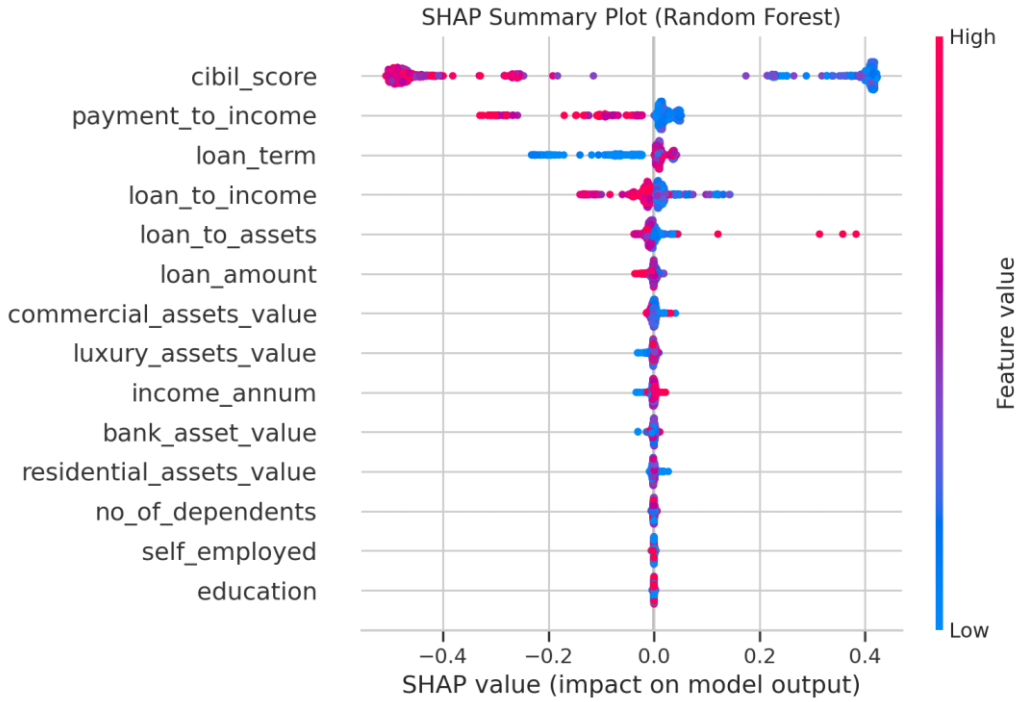


Figure 8: SHAP summary plot of the Random Forest model on the test set

Interpretability with SHAP. SHAP assigns every feature a contribution to each individual prediction, showing how much it pushed the decision towards rejection or approval. The summary plot (Figure 8, computed on the Random Forest, the best single model) shows that cibil_score dominates with a mean absolute SHAP value of 0.43, far ahead of the engineered payment_to_income ratio (0.04), loan_term (0.03), the engineered loan_to_income ratio (0.02) and the engineered loan_to_assets ratio (0.01). Low CIBIL scores push the prediction strongly towards rejection and high scores push it towards approval, while education, self-employment and the number of dependents have almost no effect.

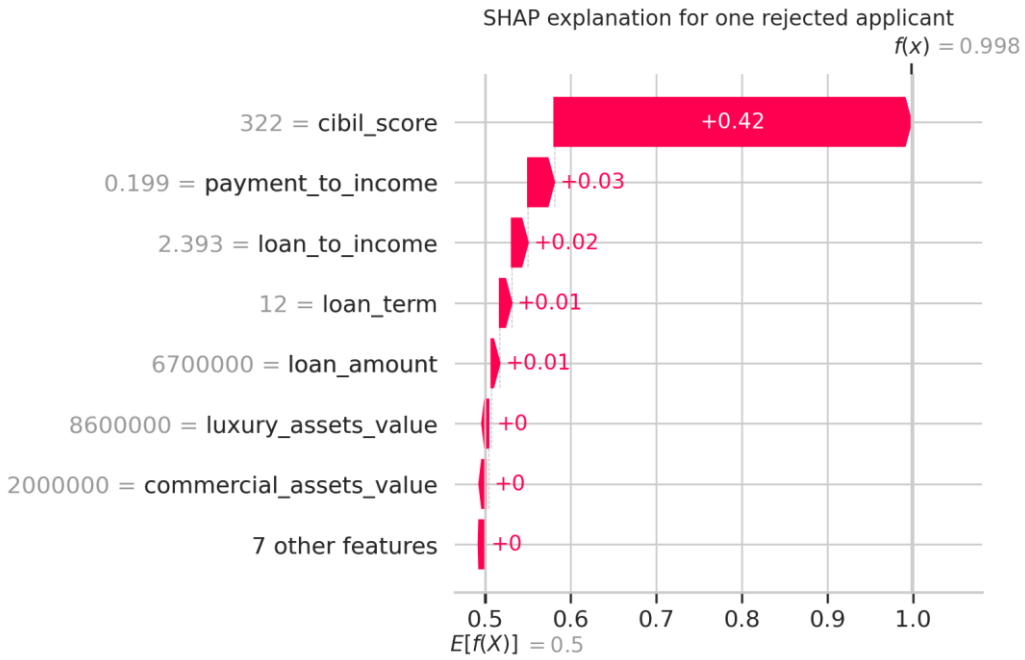


Figure 9: SHAP explanation of the decision for one applicant

Figure 9 explains a single rejected test applicant (annual income 2.8 million, loan amount 6.7 million, 12-year term, CIBIL score 322). The model gives a 99.8% probability of high risk, and the low CIBIL score alone contributes +0.42, with smaller contributions from payment_to_income (+0.03), loan_to_income (+0.02) and loan_term (+0.01). This kind of applicant-level explanation can be shown to the applicant or a regulator to justify the decision.

Conclusion

This case study demonstrated the complete machine-learning workflow for predicting loan approval from applicant financial data: cleaning, EDA, feature engineering, pre-processing, handling of class imbalance, training of three tree-based classifiers, soft-voting ensembling, evaluation and interpretation. The EDA showed that the CIBIL score is by far the attribute most associated with the decision (correlation −0.77), while education, self-employment, dependents and asset values have almost no effect. The soft-voting ensemble of Random Forest, XGBoost and LightGBM classified all 854 test applications correctly (accuracy 100%, F2-score 1.000, ROC-AUC 1.000), matching Random Forest and removing the single errors made by XGBoost and LightGBM. SHAP analysis made the decisions explainable, both globally and for an individual applicant.

The study has limitations: a small dataset that is almost fully separable by one feature, which explains the near-perfect scores; a single train-test split supported by cross-validation on the training data only; fixed, untuned hyper-parameters; ratios that only approximate debt-to-income and credit utilisation because the dataset has no debt or credit-limit columns; and the possibility that historical decisions contain bias. Future work can include systematic hyper-parameter tuning, cost-sensitive threshold optimisation based on actual default losses, fairness audits across applicant groups, stacking ensembles, and validation on an independent and more realistic dataset that contains real debt and credit-utilisation data.

In conclusion, the results suggest that an explainable ensemble of bagging and gradient-boosting models can provide useful decision support for credit assessment, but such a model should only supplement, and never replace, the judgement of qualified credit professionals.

Signature of the Team members